Senior DevOps Engineer Skills Assessment

Expected time: 2–3 hours
Format: Take-home case study
Tools: Internet, documentation, and AI tools are explicitly allowed.

Scenario

You recently joined TechnologyAdvice as a Senior DevOps Engineer. You support a distributed Engineering organization running production applications primarily in AWS using Kubernetes, Infrastructure as Code, GitOps/CI/CD, and modern observability tooling.

Over your first few weeks, you identify several recurring operational problems:

Production Reliability

A customer-facing application has experienced three significant incidents in the past two months. In the most recent incident, application pods repeatedly restarted because they were running out of memory. Kubernetes restarted the containers, but customer requests experienced elevated errors for approximately 35 minutes.

The team received an alert after customer errors had already increased significantly. An engineer manually increased the application's memory allocation, which restored service.

There has been no permanent follow-up beyond increasing the memory limit.

Deployment Process

Engineers deploy through GitHub Actions and ArgoCD. Most deployments are automated, but production releases still require several manual steps documented in an internal runbook.

Engineers frequently ask DevOps for help when deployments fail, and the DevOps team spends approximately **6–8 hours per week** assisting with deployment-related issues.

Environment Provisioning

Creating infrastructure for a new service generally requires a DevOps Engineer to copy and modify Terraform from an existing service.

Provisioning a new environment typically takes **1–2 business days**, depending on DevOps availability.

Cloud Cost

AWS infrastructure spend has increased approximately **25% over the last six months**, while application traffic has increased approximately **8%**.

No one has yet determined what is responsible for the difference.

Observability

Teams use Datadog, but monitoring has evolved organically. There are many alerts, engineers report alert fatigue, and several recent production incidents were first noticed through customer reports rather than monitoring.

Leadership is not asking you to fix everything immediately.

Your responsibility is to determine **where you would focus first and how you would improve the environment over time.**

Please complete the following 6 deliverables:

Rank the problems you would address and explain why.

You do not need to solve everything. Explain what you would investigate or address first, what you would intentionally defer, and what additional information you would gather before making those decisions.

During the most recent application incident:

- Memory utilization increased gradually for approximately 90 minutes.
- Pods eventually reached their memory limits and were OOMKilled.
- Kubernetes restarted the affected pods.
- Application error rate increased significantly during the restart cycle.
- The existing memory alert triggered after the application was already experiencing elevated errors.
- An engineer doubled the memory limit, and the application stabilized.
- No application deployment occurred immediately before the incident.

Walk us through how you would investigate this incident. What hypotheses would you explore, what data would you want, and what improvements might you consider after determining root cause?

Choose ONE recurring operational problem from the scenario and propose a systemic improvement that would reduce or eliminate the need for repeated DevOps intervention.

Describe:

- Current problem
- Proposed solution
- How it would work
- Risks/trade-offs
- How you would roll it out safely
- How you would measure whether it worked

Provide one technical artifact that demonstrates how you would begin implementing or communicating your proposed improvement.

Examples include:

- Architecture diagram
- Terraform/IaC example
- GitHub Actions workflow
- Kubernetes configuration
- Automation script
- Monitoring/alerting design
- Runbook
- Pseudocode
- Platform/self-service design

Production-ready code is **not required**.

You are encouraged to use AI tools during this assessment. We are interested in how you use them, not whether you use them.

Briefly describe:

- Where you used AI during the assessment
- What you asked it to help with
- One recommendation/output you accepted or found useful
- One output you challenged, rejected, corrected, or independently verified
- How you would validate AI-generated infrastructure or operational recommendations before applying them to production

---

Write a short Slack update (maximum 150 words) to the VP of Technology explaining the recent production incident, what is currently known, what is not yet known, and what happens next. Assume the VP is technical but does not need low-level implementation details.